

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 2.1: Reproduction in Plants — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.1: Reproduction in Plants
Driving Question	DRIVING QUESTION: Why do some farm plants produce offspring without seeds while others require seeds and a partner tree to make fruit?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Cuttings, Tubers and Lonely Trees: Why Do Plants Reproduce So Differently?	What did you observe on the sweet potato tuber, onion bulb, ginger rhizome, grass runner and bryophyllum leaf? Record the structure and the place where a new plant could start (buds, nodes, leaf margin).	What did you learn about how plants such as cassava, sugarcane, sweet potato, ginger and onion can reproduce without seeds? Include the idea of one parent, stored food and identical offspring, and name at least one advantage and one disadvantage.	How does this evidence help to explain part of the driving question, and what is still unexplained? Why might maize and beans need seeds, and why might a lone avocado or pawpaw tree full of flowers not set fruit? Write the new question that leads into Lesson 2.
Lesson 2 Flower Detectives: Classifying Flowers by Ovary Position, Symmetry, Inflorescence and Sex	We dissected Hibiscus, Bougainvillea, maize inflorescence and passion flower. We saw that Hibiscus is bisexual, with stamens and carpel in one flower; maize has separate male (tassel) and female (silk/cob) flowers on the same plant; Bougainvillea has small flowers surrounded by showy bracts; and flowers differ in symmetry, ovary position and arrangement in clusters.	Flowers can be classified by ovary position (superior or inferior), symmetry (actinomorphic or zygomorphic), inflorescence (single or clustered) and by whether they are bisexual or unisexual. The ovary holds the ovules, and its structure shows how the seeds are attached (placentation).	A lone pawpaw or avocado tree may flower heavily but not fruit because fruit needs pollen from stamens to reach the ovary of a carpel. If the tree's flowers are unisexual or lack the matching partner parts, pollen must come from another flower or tree. Cassava and sweet potato cuttings skip flowers altogether. New question: if flowers have both male and female parts, why don't all flowers pollinate themselves?
Lesson 3 Why Can a Lone Pawpaw Tree Flower Heavily but Never Set Fruit? Adaptations That Prevent Self-Pollination	We used wire/bead models and flower charts to trace pollen movement in maize, Hibiscus, pawpaw and passion fruit. We saw that pollen and stigma can be ready at different times (protandry/protogyny), that styles and stamens can be at different heights (heterostyly), that some trees have	Many flowering plants have timing, structural, separate-sex and chemical adaptations that prevent self-pollination and promote cross-pollination, so a lone tree like a pawpaw or passion fruit may flower but not set fruit without a compatible partner. Plants propagated by cuttings	A lone pawpaw tree covered in flowers can stay fruitless because it is male or female only (dioecy), or because its flowers cannot pollinate themselves, so a partner tree is needed to supply compatible pollen. A cassava or sweet potato cutting grows into a new plant without flowers or a partner

	only male or only female flowers (dioecy), and that some stigmas reject their own pollen (self-incompatibility).	(cassava, sweet potato, sugarcane) bypass these barriers because no pollination is needed.	because it reproduces asexually. Next question: What actually happens inside the flower after pollen successfully lands on a receptive stigma?
Lesson 4 Following the Pollen Tube: How Double Fertilisation Explains Why a Lone Pawpaw Flowers but Sets No Fruit	We watched cached animations showing a pollen grain land on a stigma, germinate and grow a pollen tube down the style to the micropyle of the ovule, releasing two sperm nuclei into the embryo sac. We traced and labelled this pathway on our worksheets.	Double fertilisation: one sperm nucleus fuses with the egg cell to form the zygote (the future embryo), and the other fuses with the two polar nuclei to form the triploid endosperm (food store, as in maize grain for ugali). Pollination is the transfer of pollen; fertilisation is the fusion of nuclei.	A lone pawpaw or avocado tree can be full of flowers but set no fruit because without compatible pollen from a partner tree no pollen tube grows, no fusion occurs, and no zygote or endosperm forms, so no seed or fruit develops. Cassava, sugarcane and sweet potato cuttings need no fusion of gametes, so they grow into new plants without flowers or seeds. New question: How does double fertilisation transform the ovary and ovules into fruits and seeds?
Lesson 5 Inside the Fruit: Placentation as Evidence of Post-Fertilisation Development	What did you observe when you sectioned the tomato, orange, bean pod, passion fruit and cucumber? Describe where the seeds were attached and how many chambers each fruit had.	What did you learn about how fruits and seeds develop after fertilisation, and how do marginal, axile, parietal and free-central placentation differ?	Using Claim-Evidence-Reasoning, explain why a lone avocado or pawpaw tree full of flowers may not set fruit, while a cassava cutting grows into a new plant without any seed. What new question do you now have about sexual versus asexual reproduction in farming?
Lesson 6 Variation or Copies? Debating the Trade-offs of Sexual and Asexual Reproduction in Kenyan Crops	We observed from the case cards that plants grown from cuttings, suckers or setts (cassava, sugarcane, sweet potato, banana) are identical to the parent and spread quickly, but one disease like banana wilt can affect all of them. Seed-grown crops such as maize, beans and passion fruit vary from plant to plant, and fruiting trees like a lone pawpaw or avocado may not set fruit without a partner tree.	We learned that asexual reproduction needs only one parent and gives fast, identical offspring that are useful for colonising an area and keeping good traits, while sexual reproduction needs pollination and fertilisation and gives genetic variation that helps some offspring survive disease and changing conditions. Each strategy has advantages and disadvantages.	We can explain that cassava, sugarcane and sweet potato can grow from cuttings with no seed because new plants regrow from parent tissue and are clones, while maize, beans, pawpaw and avocado produce fruit and seeds through flowers, pollination and fertilisation, which creates variation and often needs another plant. A lone tree full of flowers may not fruit because it lacks a pollen source, whereas a stem piece needs no flower or partner to start a new plant.
Lesson 7 Putting the Evidence Together: Why Cassava Cuttings Need No Seeds but Lone Pawpaw and Avocado Trees Need a Partner	What we observed: a hibiscus (or tomato or passion fruit) flower dissected to show stigma, style, ovary, ovules, anthers and filaments; the distance between anthers and stigma; fruits cut open to show placentation; the animation of double fertilisation; and cassava, sugarcane and sweet potato cuttings that grow into new plants without seeds.	What we learned: asexual reproduction by cuttings and other vegetative parts gives identical plants quickly without flowers or pollination; sexual reproduction needs flowers, pollination (often cross-pollination because of barriers such as separate male and female trees, different timing and self-incompatibility), double fertilisation, and then seed and fruit development; and both methods are important for farming.	How this explains the phenomenon: a cassava cutting grows from stem tissue of the parent, so no flower or partner is needed. A lone pawpaw or avocado tree has flowers but barriers to self-pollination prevent fertilisation, so without pollen from another tree the ovules are not fertilised and no fruit or seed forms. Maize and beans must come from seeds because they reproduce sexually: pollination and double

			fertilisation produce the embryo and endosperm inside the seed.
Lesson 8 Final Explanation: Why Cassava Cuttings Grow Without Seeds but a Lone Pawpaw Needs a Partner Tree	What did we observe across the unit? Examples: cassava, sugarcane and sweet potato cuttings producing roots and shoots without flowers or seeds; hibiscus, maize, bean and tomato flower parts and how they hinder self-pollination; double fertilisation in the animation; fruit sections of passion fruit and tomato showing seeds attached by placentation; a lone pawpaw or avocado with flowers but no fruit.	What did we learn? Asexual reproduction (vegetative propagation) forms identical offspring (clones) by mitosis, is fast and needs no partner but offers no variation. Sexual reproduction needs pollination, often cross-pollination, and double fertilisation to form seeds and fruits, and it increases variation. Features such as separate sexes, different timing of anther and stigma maturity and self-incompatibility hinder self-pollination and self-fertilisation. Ovules become seeds and the ovary becomes the fruit.	How do we explain the phenomenon? Cassava, sugarcane and sweet potato cuttings regrow because the stem tissue divides by mitosis to form roots and shoots, giving a clone with no seeds needed. Maize, beans, pawpaw and avocado must reproduce sexually: a lone flowering tree may not set fruit because its flowers cannot pollinate themselves, so pollen from a partner tree is required for double fertilisation, and only then do the ovules form seeds and the ovary form fruit.

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